# Code Publication

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## Checklist overview

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## Fill out file headers

Every file in a project should have a header that allows it to be interpreted on its own. It should include the name of the project and a short description for what this file (among the many in your project) does specifically. See template here.

## Clean up comments

Make sure comments in the code are for code documentation purposes only. Do not leave comments to self in the final script files.

## Document functions

Every function you write must include a header to document its purpose, inputs, and outputs. See template for the function documentation here.

### Remove deprecated filepaths

All file paths should be defined in 0-config.R, and should be set relative to the project working directory. All absolute file paths from your local computer should be removed, and replaced with a relative path. If a third party were to re-run this analysis, if they need to download data from a separate source and change a filepath in the 0-config.R to match, make sure to specify in the README which line of 0-config.R needs to be substituted.

### Ensure project runs via bash

The project should be configured to be entirely reproducible by running a master bash script, run-project.sh, which should live at the top directory. This bash script can call other bash scripts in subfolders, if necessary. Bash scripts should use the runFileSaveLogs utility script, which is a wrapper around the Rscript command, allowing you to specify where .Rout log files are moved after the R scripts are run.

See usage and documentation here.

### Complete the README

A `README.md` should live at the top directory of the project. Include a repository status badge from repostatus.org near the top of the file to communicate whether the codebase is actively maintained, a work in progress, or an archived analysis (see Repository Status Badges). This file also includes a Project Overview and a Directory Structure, along with contributor names and the project license. See below for a template:

> **Overview**
>
> To date, coronavirus testing in the US has been extremely limited. Confirmed COVID-19 case counts underestimate the total number of infections in the population. We estimated the total COVID-19 infections – both symptomatic and asymptomatic – in the US in March 2020. We used a semi-Bayesian approach to correct for bias due to incomplete testing and imperfect test performance.
>
> **Directory structure**
>
> - 0-config.R: configuration file that sets data directories, sources base functions, and loads required libraries
> - 0-base-functions: folder containing scripts with functions used in the analysis
>   - 0-base-functions.R: R script containing general functions used across the analysis

  - 0-bias-corr-functions.R: R script containing functions used in bias correction
  - 0-bias-corr-functions-undertesting.R: R script containing functions used in bias correction to estimate the percentage of underestimation due to incomplete testing vs. imperfect test accuracy
  - 0-prior-functions.R: R script containing functions to generate priors
- 1-data: folder containing data processing scripts NOTE: some scripts are deprecated
- 2-analysis: folder containing analysis scripts. To rerun all scripts in this subdirectory, run the bash script 0-run-analysis.sh.
  - 1-obtain-priors-state.R: obtain priors for each state
  - 2-est-expected-cases-state.R: estimate expected cases in each state
  - 3-est-expected-cases-state-perf-testing.R: estimate expected cases in each state, estimate the percentage of underestimation due to incomplete testing vs. imperfect test accuracy
  - 4-obtain-testing-protocols.R: find testing protocols for each state.
  - 5-summarize-results.R: summarize results; obtain results for in text numerical results.
- 3-figure-table-scripts: folder containing figure scripts. To rerun all scripts in this subdirectory, run the bash script 0-run-figs.sh.
  - 1-fig-testing.R: creates plot of testing patterns by state over time
  - 2-fig-cases-usa-state-bar.R: creates bar plot of confirmed vs. estimated infections by state
  - 3a-fig-map-usa-state.R: creates map of confirmed vs. estimated infections by state
  - 3b-fig-map-usa-state-shiny.R: creates map of confirmed vs. estimated infections by state with search functionality by state
  - 4-fig-priors.R: creates figure with priors for US as a whole
  - 5-fig-density-usa.R: creates figure of distribution of estimated cases in the US
  - 6-table-data-quality.R: creates table of data quality grading from COVID Tracking Project

  - 7-fig-testpos.R: creates figure of the probability of testing positive among those tested by state
  - 8-fig-percent-undertesting-state.R: creates figure of the percentage of under estimation due to incomplete testing
- 4-figures: folder containing figure files.
- 5-results: folder containing analysis results objects.
- 6-sensitivity: folder containing scripts to run the sensitivity analyses

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Wu SL, Mertens A, Crider YS, Nguyen A, Pokpongkiat NN, Djajadi S, et al. Substantial underestimation of SARS-CoV-2 infection in the United States due to incomplete testing and imperfect test accuracy. medRxiv. 2020; 2020.05.12.20091744. doi:10.1101/2020.05.12.20091744

When possible, also include a description of the RDS results that are generated, detailing what data sources were used, where the script lives that creates it, and what information the RDS results hold.

## Clean up feature branches

In the remote repository on GitHub, all feature branches aside from master should be merged in and deleted. All outstanding PRs should be closed.

## Create GitHub release

Once all of these items are verified, create a tag to make a GitHub release, which will tag the repository, creating a marker at this specific point in time.

Detailed instructions: Managing releases in a repository.

## Releasing to CRAN

If your R package is intended for public distribution, you may want to release it on CRAN (the Comprehensive R Archive Network). This section summarizes the release process and adds guidance on versioning and tagging. For full details, see the R Packages book chapter on releasing to CRAN by Hadley Wickham and Jennifer Bryan.

The CRAN submission pipeline goes through several stages: initial upload, automated checks, human review by CRAN maintainers, and final acceptance or rejection. For a visual overview of the full pipeline, see the CRAN stages diagram by Edgar Ruiz.

### Versioning

R package versions follow a `major.minor.patch` scheme (e.g., `1.2.3`). Use a fourth development component (e.g., `1.2.3.9000`) on the development version to distinguish it from the released version.

- Increment the **patch** version (`1.2.3` → `1.2.4`) for bug fixes
- Increment the **minor** version (`1.2.3` → `1.3.0`) for new features that are backward-compatible
- Increment the **major** version (`1.2.3` → `2.0.0`) for breaking changes

Update the version in `DESCRIPTION` before each release.

### Pre-release Checklist

Before submitting to CRAN, work through the following steps:

1. **Update `NEWS.md`**: Document all changes since the last release in `NEWS.md`. Use `usethis::use_news_md()` to create this file if it doesn't exist. Each release should have a top-level heading (e.g., `# mypackage 1.2.0`).
2. **Run `devtools::check()`**: Ensure your package passes `R CMD CHECK` with no errors, warnings, or notes. Use `cran = TRUE` to enable the stricter checks that more closely mirror what CRAN runs. Address any issues before submitting.

   ```
   devtools::check(cran = TRUE)
   ```

3. **Check spelling**: Run `devtools::spell_check()` to catch typos in documentation.

   ```
   devtools::spell_check()
   ```

4. **Check URLs**: Run `urlchecker::url_check()` to catch broken or redirected URLs in your documentation, which CRAN flags.

```
urlchecker::url_check()
```

5. **Test on multiple platforms**: CRAN requires packages to work across operating systems and R versions. Use the following to test on additional platforms:

   - `devtools::check_win_devel()` - tests on Windows with the development version of R
   - R-hub - tests on a wide range of platforms. In rhub v2, this runs via GitHub Actions: run `rhub::rhub_setup()` once to add the workflow to your repository, then `rhub::rhub_check()` to trigger a check run. (If you previously used older rhub APIs, see the rhub v2 documentation for migration details.)

6. **Review CRAN policies**: Ensure your package complies with the CRAN Repository Policy.

### Submitting to CRAN

Once all checks pass, submit with:

```
devtools::submit_cran()
```

This bundles your package and uploads it to CRAN's submission portal. You will receive an email asking you to confirm the submission. Review time varies; watch for email from CRAN maintainers with acceptance or requested changes.

On the CRAN web submission form that `devtools::submit_cran()` opens, briefly describe any `R CMD CHECK` notes in the submission comments and explain why they are acceptable. Be concise and factual.

For first-time submissions, also check the new submission checklist in the R Packages book for additional requirements.

### After CRAN Acceptance

Once CRAN accepts your package:

1. **Tag the release and create a GitHub release**: Follow the steps in Section , using a tag named `v` followed by the version number (e.g., `v1.2.0`). Copy the relevant section of `NEWS.md` as the release notes. Tags permanently mark the exact commit that was released, making it easy to return to or compare against any released version.

2. **Bump the development version**: After releasing, append the `.9000` development suffix to the current version in `DESCRIPTION`:

```
usethis::use_version("dev")
```

   For example, after releasing `1.2.0`, the `DESCRIPTION` version becomes `1.2.0.9000`. This signals to contributors that the repository reflects development work toward the next release.

3. **Commit the version bump**: Stage and commit the updated `DESCRIPTION` (and `NEWS.md` if you added a `# mypackage (development version)` header):

```
git add DESCRIPTION NEWS.md
git commit -m "Start development toward next release (1.2.0.9000)"
git push
```

4. **Track adoption over time**: the {Visualize.CRAN.Downloads} package plots a CRAN package's download counts (retrieved via {cranlogs}), producing static, interactive, and multi-package comparison plots, with moving averages and confidence bands to make trends visible. With no date range given, it defaults to the year of downloads ending today:

```
Visualize.CRAN.Downloads::processPckg("yourpackage")
```